Department of Leather Engineering

Khulna University of Engineering & Technology

B. Sc. Engineering 4th Year 1st Term Examination-2021

Leather Products Engineering III

LE 4109

Time: 3.0 hours Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION–A

1(a)	Define leather garments. Mention the chronological summary of clothing history.	12
1(b)	What are the purposes of wearing clothes? Mention the considering factors in developing a collection.	10
1(c)	Mention popular clothing for men and women in two distinguished ancient eras Rome and Renaissance.	08
1(d)	What is the purpose of notches on a pattern?	05
2(a)	Define a blouson jacket. Mention the speciality of leather jackets.	07
2(b)	Briefly epxlain the people wear clothes from the following perspective: **(i)** Protection **(ii)** Modesty and **(iii)** Communication	12
2(c)	Compare among suit, coat and blazer.	06
2(d)	Illustrate the following terms with related figure: **(i)** Rever **(ii)** Lapel **(iii)** Grain line and **(iv)** Crown height.	10
3(a)	Mention the significance of 'Grading' and 'Enlargement'.	09
3(b)	Illustrate the enlargement techniques for blouson jacket (Back panel) with related figure.	16
3(c)	Explain –why front sleeve curve is more tapered than back sleeve curve?	10
4(a)	Draw a perspective view of a jeans skirt and specify its different parts.	07
4(b)	How will you develop an enlarged sleeve block along with body section in case of jacket?	12
4(c)	Cite the function of pleat and compare among knife pleat, inverted pleat and box pleat.	11
4(d)	Distinguish between bell skirt and puffy skirt.	05

SECTION–B

5(a)	How can you accessorize the waist coat? Show the different types of necklines and lapels used in waist coat.	12
5(b)	Show the complete set of patterns for a designer ladies waist coat. Note that the waist coat must be of single breasted and contain lapel.	11
5(c)	State the major fitting problems of a waist coat. Describe the techniques to be adopted to solve the limitations.	12

6(a)	What are the basic differences between flap pocket and besom pocket? State the following terminology: (**i**) Body rise and (**ii**) Half across back.	11
6(b)	Mention the sequence of measurement required to be taken to develop a two piece trouser block.	05
6(c)	Describe briefly the topside construction of a trouser block.	12
6(d)	Briefly explain the back preparation of a trouser.	07
7(a)	What do you mean by fitting problem? What are the common reasons behind the fitting problems in an outfit?	05
7(b)	Describe the assembling sequence of a trouser.	15
7(c)	List out the major fitting problems of a trouser. Troubleshoot the fitting problems of a trouser.	15
8(a)	What do you mean by unisex wear? Write short notes on the followings: (**i**) Princess line, (**ii**) Pelerine, (**iii**) Puddle suit.	15
8(b)	Define accessories. How do accessories differ from fashion and trimmings.	07
8(c)	Classify garment accessories. Which types of accessories can you use to make a trouser?	13

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B. Sc. Engineering 4th Year 1st Term Examination-2021

Tannery Waste Management

LE 4029

Time: 3.0 Hours | Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
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iii) Assume reasonable data if any missing.

SECTION–A

1(a) Define BOD. Differentiate between carbonaceous BOD and nitrogenous BOD. 10

1(b) Prove that, $\frac{L_t}{L_0} = e^{-kt}$ where symbols have their using meanings. 09

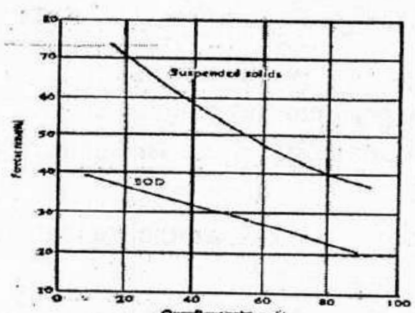
1(c) Why COD value of wastewater is higher than the BOD value? 06

1(d) BOD of a sewage incubated for 3 days at 27°C was measured 110 mg/L. Calculate BOD_5 at 20°C. Consider k = 0.23 per day (base e) and temperature coefficient = 1.047 10

2(a) Define SVI. State the role of zeta potential on coagulation process. 11

2(b) Assume, you have a wastewater mixture with higher colloidal concentration. A proper treatment is needed for discharging the water into irrigation land. Now, during the chemical treatment which coagulation mechanism will work appropriately and how? Discuss about the coagulation mechanism. 13

2(c) A municipal wastewater treatment plant processes and average flow as high as 4500 m^3/d, with a peak flows of 12000 m^3/d. Design the clarifier with a 65% of suspended solid removal. 11



3(a) What is the relationship between coagulant dosage and turbidity of wastewater? Also mention the proper coagulation mechanism on each step of coagulants dose. 13

3(b) A floating aeration system with 20 hp each is used to remove BOD and fully nitrify the waste water. The temperature of the waste water is 54°F, site elevation is 5000ft, the diffuser depth is 18 ft and there is a 2.0 mg/L dissolved oxygen residual. What are the oxygen demand, total horse power requirement, number of aerators and airflow requirement using an influent flow of 3.5 mgd, 140 mg/L BOD primary effluent, 15 mg/L in the plant effluent, and 23 mg/L ammonia? Dissolved oxygen at 5000 ft elevation is 8.8 mg/L, α = 0.65, β = 0.95 and F = 0.75 and SOTR for the floating aerators 2.0–3.6. 18

3(c)	Which type of equalization tank is appropriate for wastewater treatment plant and why?	04
4(a)	Write notes on: (**i**) Slow sand filter and (**ii**) Dual media filter	08
4(b)	Elucidate following activated sludge reactors with advantages and disadvantages: (**i**) Contact stabilization and (**ii**) Activated sludge with selector	08
4(c)	Write down the objectives of tertiary treatment for tannery wastewater. Which one is the easiest method for disinfection of water?	08
4(d)	To produce drinkable water from biologically treated tannery wastewater which treatment method could be used and how?	05
4(e)	Write notes on: (**i**) RO and (**ii**) Trickling filter	06

SECTION–B

5(a)	What is composite? State the role of reinforcement for composite material.	08
5(b)	Mention the objectives and strategies of solid waste management (SWM).	07
5(c)	Explain the environmental impact of the chemical constituents of tannery solid wastes.	14
5(d)	Describe- factors that contribute to the performance of composite.	06
6(a)	Describe the utilization of (**i**) Vegetable tanned trimming and (**ii**) Finishing trimming	10
6(b)	State the conventional model for safely disposal of tannery solid waste as landfill.	10
6(c)	Write down the major criteria of matrix for the preparation of composite material.	07
6(d)	Describe the utilization of (**i**) Buffing dust (**ii**) Wet white trimmings and shavings	08
7(a)	Explain- how can remove sulfide from sludge?	09
7(b)	Define fibre reinforced composite. Why we use leather fiber for the preparation of composite material?	08
7(c)	Write down the technology of bio-gas production from tannery solid waste.	10
7(d)	Write short notes on: (**i**) Composting and (**ii**) Incinaration	08
8(a)	What is cleaner technology? Write down the components of sustainable development in tanning industry.	07
8(b)	Write down the necessary procedures for recycle or reuse of (**i**) Used lime liquor and (**ii**) Salt from soaking liquor	10
8(c)	Discuss briefly the recovery of chromium from chrome containing effluent by chemical precipitation with parameters.	09
8(d)	Mechanical and thermal properties of leather fiber reinforced composite depend on fiber length, orientation and volume fraction-why.	09

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Footwear Engineering III

LE 4111

Time: 3.0 Hours — Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

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iii) Assume reasonable data if any missing.

SECTION–A

1(a)	Define closing. What are the points to be remembered during attaching topline tape in a shoe?	08
1(b)	What is nicking and pleating? Describe **(i)** U-bound and **(ii)** French bound topline treatments.	10
1(c)	Write down the closing sequence of Derby/Gibson shoe.	10
1(d)	Why bottom thread size is lower than top thread size?	07
2(a)	Mention some common faults in sewing department.	07
2(b)	Write down ten golden rules for closing department for shoe upper preparation.	10
2(c)	How can you control the quality of a closing department?	10
2(d)	What are the problems moisture content creates during transportation of footwear in sea root and how you can control it?	08
3(a)	Write down about the hazards of footwear industry.	10
3(b)	What is article number of a product? What do you mean by article no 854-6077?	06
3(c)	Define shoe specification sheet. Design a plant layout for 40 pair in 8 hours shoe production in a day.	12
3(d)	Write short notes on die-less cutting systems: i) Laser cutting ii) Water-jet cutting	07
4(a)	Mention some clicking faults and their effects on subsequent department.	10
4(b)	Define material allowance. Describe 180^0 method of RSM for material allowance.	10
4(c)	Suppose, you have been given 150 sdm 2nd grade leather in order to make 12 pairs of oxford shoe and the scale area for per pair of shoe is 11.50 sdm. Calculate **(i)** 2nd wastage per pair **(ii)** 3rd wastage per pair **(iii)** Total leather allowance per pair and **(iv)** How many pair of shoes are possible?	15

SECTION–B

5(a)	Write down the flow chart of operational sequences to manufacture complete footwear.	15
5(b)	"Moisture absorbance, quick dry out, and durability against friction should be the prime properties of insole materials" – Justify the statement.	12
5(c)	"Both plastic insole and blended insole are suitable for production process along with cost reduction"- Justify the statement.	08

6(a)	Why temporary attachment by adhesive is mandatory in lasting room?	08
6(b)	"It is wrong to think necessary to pull the shoe into shape by over stretching to conform to the last"- Justify the statement.	08
6(c)	State the possible cause and remedy of the following problems of toe and forepart lasting operation: **(i)** Upper torn or displaced on last **(ii)** Poorly defined feather edge **(iii)** Poor upper tension all round and **(iv)** Damaged insole	12
6(d)	Explain about the basic points to look for a successful seat and side lasting operation.	07
7(a)	Write down the complete flowchart of shoe finishing operations for synthetic footwear.	10
7(b)	"The choice of cleaner and the method of cleaning should be suitable for contamination removal and the type of upper finish"- Justify the statement.	12
7(c)	Why dry cleaning of shoe upper is more compatible than wet cleaning?	08
7(d)	Discuss the role of fillers (repairing agent) in shoe finishing.	05
8(a)	What are the factors need to be considered during top dressing operation?	08
8(b)	"Water based top dressing is more suitable for shoe upper than solvent based top dressing"- Justify the statement.	08
8(c)	What is California slip lasted shoe construction? Why it is called force lasted shoe construction?	12
8(d)	Draw the basic structure of Veldtschoen shoe construction and explain its special features.	07

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Computer Integrated Manufacturing Systems

LE 4123

Time: 3.0 Hours Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.
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SECTION–A

1(a) What is meant by CIM and CAM? Illustrate the CIM and components with figure. 09

1(b) Cite the elements of CIM hardware and CIM wheel. What are the challenges towards the implementation of CIM for industry 4.0? 07

1(c) Explain–the triology of CIM integration and its adaptation benefits for the leather products industry. 08

1(d) Clarify the terms-M_{LT} and Tp. A custom-made leather goods industry requires 7.0 days to complete 60 units of wallet. Recently, the industry has been acclimatized to automation and produced the same leather size of 60 units again which required a sequence of 09 operations in the plant. The average setup time is 3.0 hr and the avarege operation time per machine is 6.0 min. The average non-operation time due to handling, delays, inspections and so on is 7.0 hr, compute how many days it will take to produce a batch of 60 units and how do you explain the status of M_{LT}, assuming that the plant operates on a 6-hour shift per day. 11

2(a) Define custom design adaptive gripper. Write down the usual back-end algorithm for a robot function. 08

2(b) Distinguish among pitch, piece and packet regarding footwear pattern engineering. 06

2(c) "Mere introduction of robots or automation in manufacturing, may or may not contribute to the final productivity, quality or even cost". Justfy with example. 05

2(d) Mention the principle of CNC and DNC machines. 06

2(e) Figure out a 3D digitizer. Illustrate CAD for automotive leather goods. List out the sequential command of adding lap allowance along with measurements for vamp construction via power software. 10

3(a) Show the matrix of a few functional footwear with salient features. 08

3(b) What are the compulsory factors in designing a sprinting shoe? Compute and prove that the value of abridged resultant torque could be represented by the force, $\left|\overline{F_{Hy}}\right| = \dfrac{(Dx - dx)\,Nfy}{dx\,\mathrm{Sin}\,\theta}$; for a super spike sprinting footwear. 14

3(c) Advances in spike technology for running shoe designed by CADD engineering is to help run faster on certain, uneven terrain and provide extra grip to the forefoot to force forward and improve stride turnover. Usain Bolt used a Puma-spike sprinter with 6.0 no of spikes in Tokyo Olympic and this special architecture helped to reduce CTF to a certain degree and was observed 10.2 N-M. At the same time, NIKE ^{TM} introduced a super spike sprinting shoe of 5.0 no spike with 3.5 mm needle like diameter and this shoe exerted a force of 600N at heel point to CT muscle. The acute angle between the ankle joint and the X-axis was 53.2°. Determine the generated torque of NIKE^{TM} super spike sprinter; if the distance between ankle joint and heel point was 0.038 m. How much fluctuation of GRF_v would obsrve due to bottom design variation? 13

4(a) State the sequential process of 2D pattern engineering and grading in order to develop a complete set of pattern for shoe manufacturing. 15

4(b) What are the fuctions of AORD software in designing bags and other small leather goods? Point out some limitaions of shoemaster@ software. 10

4(c) Delineat "Fuss Over Scale" of conventional software. Differentiate wire-frame model of the foot from the solid model. 10

SECTION-B

5(a) Define industrial automation. Mention the role of industrial automation. 10

5(b) How does automation bring down cost in respect of cost unit and production volume? 12

5(c) Briefly explain the relationship between automation and integrated manufacturing systems. 08

5(d) Draw an industrial product life cycle. 05

6(a) Define the manufacturing system. Describe the components of the manufacturing system in respect of production machines. 13

6(b) Illustrate the following classical manufacturing systems: (i) Job shop and (ii) Contineous process 12

6(c) Established a relationship between the category of production and product variety. 10

7(a) Define material handling systems. Mention the unit load principle regarding efficient material handling systems. 07

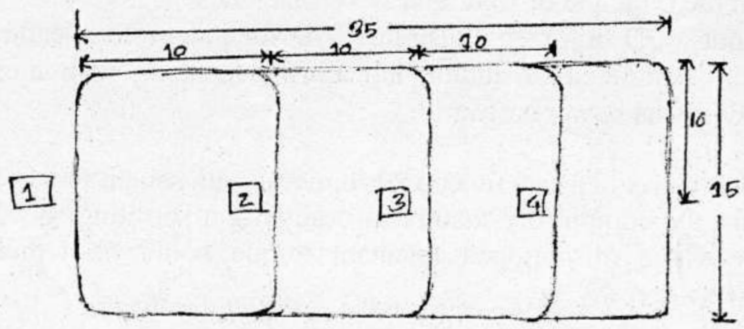
7(b) What are Automated Guided Vehicle (AGV) systems? Briefly explain the different types of AGVs systems in the manufacturing industries. 10

7(c) A flexible manufacturing system (FMS) is being planned. It has a ladder layout as picture in Fig. It uses a rail guided vehicle (RGV) system to move parts between situations in the layout. All work parts are loaded into the system at station 1, move to 1 of three processing stations (2, 3 or 4) and then brought back to station 1 for unloading. Once loaded on to its RGV, each workpart stays on board the vehicle throughout its time the FMS. Load and unload times at staion 1 are each 1.0 min. Processing time are: 5.0 min at station 2; 7.0 min at station 3; and 9 min at station 4. Hourly production of parts through the system is : 7 parts through station 2; 6 parts through station 3; and 5 parts through station 4. Now determine- 12

(i) Develop the from-to chart for trips and distances;

(ii) Develop the network diagram

(iii) Determine the number of rail guided vehicle that are needed to meet the requiremnts of the flexible manufacturing system, if vehicle speed=60 m/min and the anticipated traffic factor=0.85. assume reliability= 100%



7(d) Mention the application areas of AGVs system. 06

8(a) Define Group Technolgy (GT). Cite the salient pros and Cons of implementing the GT system. 10

8(b) Briefly describe the Optiz classification system with example. 12

8(c) Apply the rank order clustering technique to the part-machine incidence matrix in the following table to identify logical part families and machine groups. Parts are identified by letters and machines are inedentified numerically: 13

Machine	A	B	C	D	E
1	1				
2		1			1
3	1			1	
4		1	1		
5				1	